Title Page

Manuscript title:
Counterfactual Attribution of Crypto Systemic Stress on a Ledger-Specified Causal Graph: Methods and a Cautionary Benchmark

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Author's Information

Quang-Vinh Dang. Quang-Vinh Dang is a Lecturer at British University Vietnam and Data/AI Director at SmartOSC, where he leads initiatives in multi-agent AI systems, large language models, and graph neural networks across banking, retail, and finance. He holds a Ph.D. in Computer Science from Université de Lorraine (LORIA), France (2018), an M.Sc. in Financial Engineering from WorldQuant University (2021), and completed a Post-Master programme in Software Technology at Eindhoven University of Technology. His research spans machine learning in finance, natural language processing, intrusion detection, and federated learning, and he has authored more than 60 peer-reviewed papers indexed in Scopus/SCI-E, many in Q1 journals. Prior to his current roles, he held data-leadership positions at Trusting Social, Be Group, GoTymeX, and DatVietVAC, and was a postdoctoral researcher at Inria, France.

Cover Letter

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The Editor-in-Chief
Digital Finance

Dear Editor-in-Chief,

I am submitting the manuscript **"Counterfactual Attribution of Crypto Systemic Stress on a Ledger-Specified Causal Graph: Methods and a Cautionary Benchmark"** for

consideration in *Digital Finance*. Measures of crypto systemic risk usually infer contagion networks from price co-movement, using CoVaR-type or expected-shortfall statistics. The resulting graphs are estimated rather than observed, and the attributions built on them cannot be checked against the transfers that moved funds between entities. Public blockchains record those transfers, but the raw ledger becomes an entity-level graph only after addresses are resolved to institutions. The October 2025 liquidation cascade made this gap concrete.

The manuscript treats the entity-resolved on-chain ledger as the candidate parent set of a linear structural causal model of entity stress. Its methodological contribution is a framework that is analytic throughout: an interventional tail measure (On-Chain Causal CoVaR) and a counterfactual loss attribution computed by full abduction, with flow-free network-multiplier bounds, abduction invariance of stress deviations, a share bound, a walk representation of the linear Shapley attribution with an exact $O(2^p)$ subset algorithm, and a certified ranking radius. The paper also shows that the choice of abduction changes the answer. On the same data, zero-shock abduction inflates the Binance-to-Bybit share, so attributions built this way need full abduction to be interpretable.

The paper is also a cautionary benchmark, and I present it as one. Out of sample, the network equation explains little hourly stress: the pooled out-of-sample R^2 is 0.005, against 0.161 for an own-lag model. No channel's share exceeds 15.2%. H1, that the observed graph predicts better than a price-estimated graph, is not supported, and edge direction is not distinguishable from reversal. Placebo evidence is mixed. I argue that this is the value of the paper for supervisors and risk teams: a ledger-specified graph is auditable, and auditability is not the same as explanatory power.

The paper fits the scope of *Digital Finance* on three counts. It addresses systemic risk in crypto-asset markets and how that risk is measured. It addresses supervisory transparency, since each attribution traces back to public ledger records and an explicit model. It also provides reproducible inputs: the code, the query definitions, and the derived result files are openly available, and the core data come from free public sources.

This manuscript is original, has not been published previously, and is not under consideration at any other journal. It is single-authored, and the author declares no competing interests. The code and derived results are publicly available at https://github.com/vinhqdang/VEIN-On-Chain-Causal-Risk-Attribution-for-Cryptocurrency-Systemic-Risk. Thank you for considering this manuscript.

Sincerely,

Quang-Vinh Dang
British University Vietnam

Declarations

Ethical Approval and Consent to participate. Not applicable: this study uses only public, aggregated on-chain and market data and involves no human subjects, human data, or animals.

Consent for publication. Not applicable.

Availability of supporting data. All code, query definitions and derived result files underlying every number, table, and figure are publicly available at https://github.com/vinhqdang/VEIN-On-Chain-Causal-Risk-Attribution-for-Cryptocurrency-Systemic-Risk. Raw transfers are regenerated from the public BigQuery dataset `bigquery-public-data.crypto_ethereum` (token transfers), which can be queried in the free BigQuery sandbox. Address labels come from the eth-labels and etherscan-labels repositories, both MIT-licensed, and a label table built from these two sources alone is included. Prices are from CoinGecko and protocol TVL from DefiLlama. No restricted or proprietary data were used.

Competing interests. The author declares no competing interests.

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Authors' contributions. Q.-V.D. conceived the study, developed the VEIN framework and its implementation, performed the data collection and empirical analysis, and wrote the manuscript.

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Authors' information (Optional). See "Author's Information" above.